Labor

7} Palpation of Uterine Contraction

Goal:

To provide nursing students with the knowledge and skills necessary to assess uterine contractions effectively during labor, ensuring accurate evaluation and timely intervention for maternal and fetal well-being.

Learning objectives: -

At the end of this skill, students will be able to: -

1. Identify the palpation of uterine contraction.
2. List the indication of palpation of uterine contraction.
3. Analyze the appropriate time to palpate uterine contractions.
4. Recognize various methods for assessing uterine contractions.
5. Describe the purpose of palpation of uterine contraction.
6. Demonstrate the procedure of palpation of uterine contraction.

Outline:-

1. Introduction.
2. Methods of uterine contraction assessment.
3. Definition of palpation of uterine contraction.
4. Objective of palpation of uterine contraction
5. Indications of palpation of uterine contraction.
6. Normal characteristics of uterine contraction.
7. Timing of palpation of uterine contraction.
8. The procedure of palpation of uterine contraction.

Introduction:

Normal labor and delivery are dependent on the presence of regular and effective contractions of the uterine myometrium. Monitoring uterine activity during term labor and in suspected preterm labor is an important component of obstetrical care because cases of inadequate and excessive uterine activity may lead to substantial maternal and neonatal morbidity and mortality. Indeed, it is still an effective method of evaluating uterine contractions, particularly in low-resource settings.

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Definition of uterine contraction Palpation:

- ☞ It is an evaluation of the uterine efforts during labor which in turn affect cervical dilatation and expulsion of the fetus.
- ☞ Palpation has the advantage of being readily available and inexpensive. It does not require any equipment and can be easily taught.

Objectives:

1- To evaluate the uterine effort.

2- To determine the progress of labor.

3- To detect any irregularity or abnormality in labor process.

4- To reassure the woman and her family.

Integration with Fetal Heart Rate Monitoring:

Palpation of uterine contractions is also useful for determining the best time to monitor the fetal heart rate (FHR). **The optimal time** to auscultate the FHR is **immediately after a contraction**, to assess how the fetus recovers from the stress of uterine activity.

When uterine contractions are palpated: -

Palpation of uterine contractions is made *every hour* in the latent phase of labor, and *every half-hour* in the active phase or with oxytocin.

Indications of palpation of uterine contraction: -

1- To assess progress of labor in relation to length, strength and frequency of uterine contractions.

2- Correct administration of oxytocin.

3- During the 2nd stage of labor; if the woman is being instructed to push.

4- During the 3rd stage of labor; to determine whether the uterus is contracted prior to undertaking controlled cord traction to deliver the placenta and membranes.

Methods for uterine contraction assessment:

Current methods in clinical practice for assessing uterine activity are

☞ **External methods: -**

1. Manual palpation (Palpation of uterine contraction)
2. External tocodynamometry.

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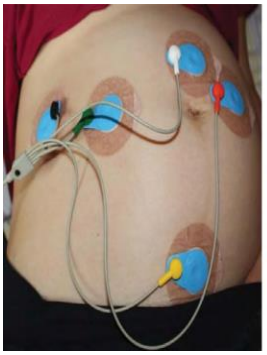
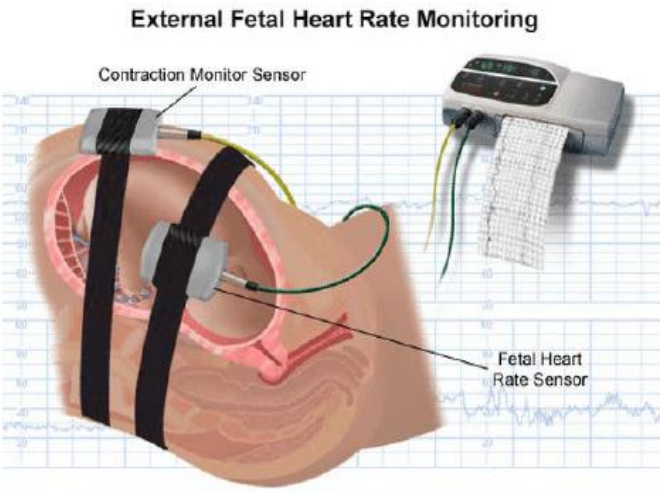


EUM (Electrical Uterine Monitoring)



- **Internal method:** IUPC (Intrauterine Pressure Catheter).

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Precautions and Contraindications of Palpation:

While palpation is safe in most cases, it should be avoided or used cautiously in the following conditions:

- Complete placenta previa.
- Unexplained vaginal bleeding.
- Multiple pregnancies with low-lying placenta.
- In high-risk cases Electronic monitoring methods such as Cardiotocography (CTG) or Intrauterine Pressure Catheter (IUPC) are preferred.

Normal characteristic of uterine contraction: -

To describe the uterine contractions, certain definitions must be recognized: the frequency; the duration; the interval and the strength.

- **The frequency:** is assessed by the number of uterine contractions in 10 minutes period. Describe per ***(time***) the *normal is (3-5 time/10 minute)*
- **The duration:** the duration of the contractions is the time, from beginning of contractions (first felt abdominally), to the time when

the contraction passes off. Described in seconds. Maximum duration: 60 seconds

- **The interval:** it is the time from the beginning of one contraction to the beginning of the following one.
  - Described in minutes.
  - Minimum interval: one and a half minutes
- **The strength or intensity:** is the rise in the intrauterine pressure produced by uterine contraction.

Described according to duration:

- ☞ less than 20 sec=mild
- ☞ 20 to 40sec=moderate
- ☞ 40 to 60 sec=severe

- **Resting Tone:** Resting tone refers to the uterine relaxation between contractions. Monitoring the resting tone is essential because a uterus that does not properly relax between contractions can lead to decreased oxygenation to the fetus.

Normal: The uterus should feel soft and easily compressible between contractions.

Abnormal: A firm or continuously hard uterus may indicate uterine hypertonus or conditions like placental abruption.

Equipment

1- A watch / clock with a second hand

2- Pencil and paper.

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Nursing action	First		Second	
	Done	Not Done	Done	Not Done
1- Review the mother's admission history to determine the onset, frequency, duration, and intensity of contractions.				
2- Greet the mother, Introduce yourself and explain the procedure.				
3- Gain informed consent				
4- Prepare equipment & take to bedside.				
5- Keep privacy& screen the mother bed.				
6- Wash hands and keep it warm				
7- Put the woman in a comfortable position (dorsal recumbent)				
8- Drape the woman and expose her abdomen only.				
9- Locate the fundus and place the hands gently on it.				
10 Assess at least three contractions in row but preferably more.				
11-Guidelines for minimum frequency of assessment are; Hourly during latent phase. Every 30 minutes during active phase. Every 15 minutes during 2nd stage.				
12-Assess more frequently if abnormalities.				

13- Assess: - Intensity (Mild, Moderate, Firm) - Frequency (in minutes & fraction) - Duration of contractions (in seconds). - Relaxation time. Interval				
14- Observe any irregularity or abnormality of contractions. Ex : hypertonic contraction - Duration longer than 90 seconds. - Interval shorter than 60 seconds. - Incomplete relaxation between contractions.				
15-Determine how the woman copes with discomfort by assessing her breathing and relaxation techniques.				
16- Discuss the findings with the woman				
17- Drape the exposed abdomen				
18- Record the following: a- Number of contractions in 10 minutes. b- Duration of contractions by seconds. c- Interval of contractions by minutes. d- Periods of relaxation by minutes. e- Intensity of contraction as mild, moderate and strong. f- Possible woman's tolerance.				
19- Remove & screen equipment				
20- Wash hands.				
Procedure score				
Date				

Student name: **Instructor name:**